THE TYPOLOGY OF INDO-EUROPEAN

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ROADMAP

- Typology and Indo-European
  - *Phonology*
    - The glottalic model
    - Apophony and laryngeals
  - *Morphology*
  - *Syntax*
    - Word order
- Gender
- Configurationality
- Basic valency and valency orientation

LANGUAGE TYPOLOGY AND INDO-EUROPEAN LINGUISTICS

Comparing languages

Early comparison: The underlying idea was that the cross-linguistic comparison of very different languages may be able to uncover the general philosophical principles … and, at the same time, recover the evolution of man's faculty of language (Ramat 2010: 32; see Jasanoff 2017: 138-238)

- Nicolas Beauzée Grammaire générale ou exposition raisonnée des éléments nécessaires du langage, pour servir defondement à l'étude des toutes les langues (1767).
- Johann Christoph Adelung / Johann Severin Vater (1806) Mithridates, oder allgemeine Sprachen-kunde mit dem Vater Unser als Sprachprobe in nahe fünfhundert Sprachen und Mundarten

Systematic comparison → end of 18th – beginning of 19th century

- The birth of Indo-European linguistics: Sir William Jones' Third Anniversary Discourse to the Asiatic Society (1786)

LANGUAGE TYPOLOGY AND INDO-EUROPEAN LINGUISTICS

August Wilhelm Schlegel (1818):

- no grammatical structure → Chinese
- agglutinated affixes → Turkish
- inflections → Indo-European
  - *synthetic (Latin, Greek)*
  - *analytic languages (Romance)*

⇩

Systematic connection between language type and language family

LANGUAGE TYPOLOGY AND INDO-EUROPEAN LINGUISTICS

Franz Bopp 1816. *Über das Conjugationssystem der Sanskritsprache in Vergleichung mit jenem der griechischen, lateinischen, persischen, und germanischen Sprache.*

Agglutinationstheorie → grammaticalization theory

Wilhelm von Humboldt 1836. *Über die Verschiedenheit des menschlichen Sprachbaues und ihren Einfluss auf die geistige Entwicklung des Menschengeschlechts.*

Classification of morphological techinques

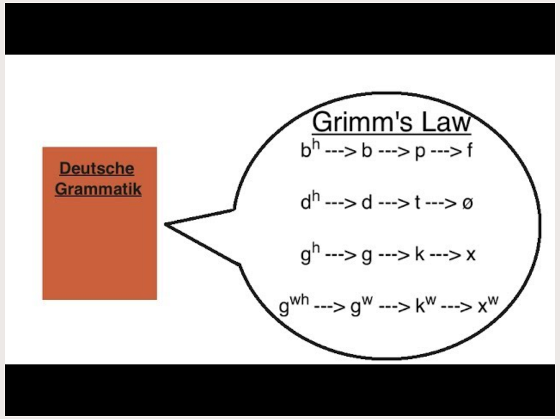
- Both works are important for morphological typology → first level of linguistic analysis to attract interest in language comparison

PHONOLOGY

The shift to phonology

Jacob Grimm *Deutsche Grammatik* (Göttingen, 1819, 2nd ed., Göttingen, 1822–1840)

600 pages devoted to phonology --> Grimm's law or First Germanic Sound Shift (Collinge 1985: 64-76)



Verner's Law (Karl Verner 1875)

Indo-European	Germanic	
/p/ /t/ /k/	[f] [θ] [x]	Grimm's Law
	[β] [ð] [ɣ]	Verner's Law

PHONOLOGY

PIE stops – Traditional reconstruction

	labial	dental	dorsal		
			palatal	velar	(labio)velar
voiceless stops	p	t	k’	k	k^w
voiced stops	b	d	g’	g	g^w
voiced asp. stops	b^h	d^h	$g’^h$	g^h	g^{wh}

PHONOLOGY

Is this system plausible?

- Languages that preserve aspirate stops either have only voiceless aspirates, as Ancient Greek or have both voiced and voiceless aspirates, as Indo-Aryan (no typological parallels for traditional PIE stops):

  Ancient Greek stops: /p/ /b/ /pʰ/

  Old Indo-Aryan stops: /p/ /pʰ/ /b/ /bʰ/

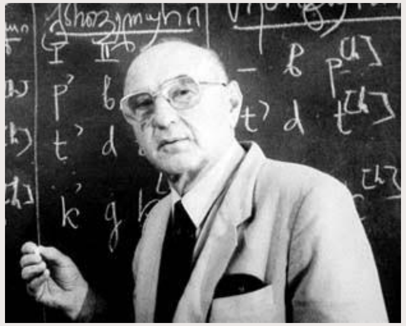
- **b* excedingly infrequent (**bh* very frequent!)
- Constrains on the distribution of stops:
  - *No sequences of two plain voiced stops (*deg impossible).*
  - *No root with both a voiceless stop and a voiced aspirate (*dʰek or *tegʰ impossible)*
  - *Plain voiced stops compatible with either of the other two series: (*degʰ or *dek both possible)*

PHONOLOGY

Jakobson, Roman 1958. Typological studies and their contribution to historical comparative linguistics. Proceedings of the Eighth International Congress of Linguists, pp. 17–35. Oslo University Press.

The reconstructed system of PIE stops had no typological parallels in any known language !!

PHONOLOGY



The glottalic theory (New look of PIE)

Gamkrelidze, Thomas V. and Vjaceslav V. Ivanov. 1984 as Indoevropeiskij jazyk i indoevropeitsi. Tbilisi: Izdatel'stvo Tbilisskogo Universiteta. (English translation 1995. Indo-European and the Indo-Europeans, 2 vols. Berlin: Mouton de Gruyter)

Hopper, Paul. 1973. Glottalized and murmured occlusives in Indo-European. Glotta 7: 141–66.

PHONOLOGY

The glottalic theory (New look of PIE)

- Voiceless stops with an aspirated allophone corresponding to the voiceless stops of the traditional reconstruction
- Glottalized stops corresponding to the voiced stops of the traditional reconstruction
- Voiced (or murmured) stops with an aspirated allophone corresponding to the voiced aspirates stop of the traditional reconstruction

a. Traditional PIE

*[p]	*[b]	*[b^h]

b. New look

*[p]/[p^h]	*[p']	*[b∠]/[b∠^{h}]

PHONOLOGY

The glottalic theory (New look of PIE)

	labials	dentals	velars[b]	labialized velars	uvulars
voiceless stops	$p \sim p^h$	$t \sim t^h$	$k \sim k^h$	$k^w \sim k^{wh}$	$q \sim q^h$
ejective or glottalized stops	(p’)	t’	k’	k^{w}’	q’
voiced stops	$b \sim b^ɦ$	$d \sim d^ɦ$	$g \sim g^ɦ$	$g^w \sim g^{wɦ}$	$ɢ \sim ɢ^ɦ$

PHONOLOGY

Typological plausibility

- similar to consonant inventories of other known languages
- glottalized stop [p’] in the place of the voiced stop [b] → this sound is often missing in phonological systems that feature glottalized stops
- it is also often the case that two glottalized stops cannot occur in close sequence

PHONOLOGY

Especially fit for Germanic!

New Look of Grimm's Law

**p/pʰ *t/tʰ *k/kʰ* the aspirated allomorph > Proto-Germanic ɸ θ x (easier than if we only postulate a non-aspirated voiceless stop) → typologically similar to one of the more notable evolutions from Ancient to Modern Greek (with φ, θ, and χ respectively).

**b/bʰ *d/dʰ *g/gʰ* → non-aspirated allomorph phonologized > Proto-Germanic b d g

**p' t' k'* → glottalization is lost > Proto-Germanic p t k

PHONOLOGY

Is the traditional reconstruction really implausible?

- Bario Kelabit (Austronesian, Blust 2006)
- Mbatto (Kwa, Stewart 1989)

Comrie (1993: 82) so-called voiced aspirates are breathy voiced stops → traditional definition of voiced aspirates for PIE ‘phonetically untenable’

- Indo-Aryan languages, which feature breathy voiced stops, either also have voiceless aspirates (Sanskrit), or have lost aspiration altogether (Sinhala). Panjabi has lost breathy voiced stops, similar to Ancient Greek. Malayam (Dravidian but heavily influenced by Indo-Aryan) breathy voiced stops have merged either with plain voiced stops or with voiceless aspirates.

PHONOLOGY

Apophony

Connected with (pre)PIE accent

PIE (possibly (pre)PIE) accent → Lexical accent "a word's accent is not determined by its purely phonological properties (such as syllable weight or metrical structure), but is rather dependent on what morphemes it contains and how they are combined." (Lundquist & Yates, 2018: 2122) (attested in e.g. Salishan, Tokyo Japanese, Chamorro (Austronesian), Cupeño (Uto-Aztecan)

Apophony visible in paradigmatic classes

↓

Kiparsky (2010) no clear parallel for the pre-PIE system has yet been brought forward.

PHONOLOGY

Apophony

Strong vs. weak verbs → Grimm (*Deutsche Grammatik*)

Strong verbs inflect through alteration of their root vowel, weak verbs inflect through the ‘help’ of an affix → strong verbs are more ‘powerful’, weak verbs show signs of ‘decay’

Some morphological types are more ‘valuable’ than others (see morphology)

PHONOLOGY

Internal consistency: Laryngeals

Ferdinand de Saussure 1879 *Mémoire sur le système primitif des voyelles dans les langues indo-européennes*

Confirmed by Hittite (known form 1916)!

After vowels

	PIE	Latin	Sanskrit	Greek	Hittite
*iH > *ī	*$g^{w}ih_2$-wós	vīvus	jīva	bíos	
*uH > *ū	*$d^{h}weh_2$-	fūmus	dhūma	thūmós	tuwaḫḫaš
*oH > *ō	*só$h_2$wl̥	sōl	sū́rya	hḗlios	
*eh_1 > *ē	*séh_1-mn̥	sēmen		hêma	
*eh_2 > *ā	*peh_2-(s)-	pāscere (pastus)	pā́ti	patéomai	paḫḫas
*eh_3 > *ō	*deh_3-r/n	dōnum	dāna	dôron	

Before vowels

	PIE	Latin	Sanskrit	Greek	Hittite
*Hi > *i	*$h_1$íteros	iterum	ítara		
*Hu > *u	*pél$h_1$us	plūs	purú-	polús	
*Ho > *o	*$h_2$owi-	ovis	ávi	ó(w)is	Luw. ḫawa
*h_1e > *e	*$h_1$ésti	est	ásti	ésti	ēšzi
*h_2e > *a	*$h_2$ent *$h_2$erǵ-	ante argentum	ánti árjuna	antí árguron	ḫanti ḫarki
*h_3e > *o	*$h_3$érb^{h}-	orbus	arbhas	orphanós	ḫarp-

PHONOLOGY

Laryngeals and typology

The phonological influence of the laryngeals on the coloring of an adjacent vowel must have been due to phonological features of the laryngeals themselves. They obviously had posterior articulation, more posterior than the uvular consonants. On typological grounds, these phonemes can be regarded as pharyngeal continuants.

This three-member set of laryngeals provides a structural parallel to the three velar orders - velar, palatovelar, and labiovelar … This system not only has typological parallels in Abkhaz-Adyghe (AlIen 1956) and modern Eastern Iranian dialects (Edel'man 1973), but in addition is justified by internal structural correlations within the Indo-European sound system.

Gamkrelidze & Ivanon (1995: 137)

PHONOLOGY

What is the present incorporation in IE linguistics / reconstruction of PIE?

- Laryngeal theory commonly accepted → based on actual shortcomings of the reconstruction
- Glottalic theory mostly not accepted → entirely based on typological considerations

MORPHOLOGY

Morphological techniques

- **Fusional** → a single inflectional morpheme to denote multiple grammatical meanings, different inflectional classes (e.g. case and number).

Latin: *amic-o* (dat.sg) *vs. amic-is* (dat.pl); *homin-i* (dat.sg) *vs homin-ibus* (dat.pl)

- **Agglutinative** → each morpheme denotes a single grammatical meaning

Turkish: *arkadaş-a* (dat.sg) *vs. arkadaş-lar-a* (dat.pl)

- **Isolating** → no inflectional morphology
- **Polysynthetic** → arguments incorporated into the verb

MORPHOLOGY

Morphological techniques

Humboldt (1836): Languages and in general language families belong to a type and cannot change.

'Inner form of a language': The character and structure of a language expresses the inner life and knowledge of its speakers … languages must differ from one another in the same way and to the same degree as those who use them.

It is the task of the morphology … to distinguish the various ways in which languages differ from each other as regards their inner form, and to classify and arrange them accordingly (*Encyclopædia Britannica*)

MORPHOLOGY

Morphological techniques

Humboldt 1836

Value scale: isolating > *agglutinative* > *fusional*

PROBLEMS:

- Chinese is isolating, but it is he language of a culturally highly evoluted society.
- Disappearance of inflection in modern IE languages of Europe (e.g. English)

Loss of inflection then becomes the highest stage of development!

MORPHOLOGY

PIE: highly fusional

- New grammatical forms due to loss of inflcetion

Schlegel 1818: “The invention which produced analytic grammars is somehow a negative one, and the method uniformly applied in this respect may be reduced to a single principle. one strips certain words of their meaningful energy, leaving them only a nominal value, in order to grant them a more general circulation and to introduce them into the elementary part of language. These words become a sort of paper money, meant to facilitate circulation.”

- Synthetic vs. analytic languages
- Definite/indefinite article and auxiliaries (including future) in Romance languages as compared to Latin

MORPHOLOGY

PIE: highly fusional

- The origin of grammatical forms

↓

Agglutinationstheorie

Franz Bopp 1816

Sanskrit: original state of affairs in IE *vs.* Changes displayed by all of the other IE languages

↓

The “simple language organism” of Sanskrit is gradually destroyed and replaced by “mechanical compositions, from which, when their elements were no longer recognized, an appearance of a new organism evolved”

→ BUT: also applies the same analysis to certain inflectional categories of this language which strike him as analyzable (Lehmann 2016: 106)

MORPHOLOGY

PIE: highly fusional

Agglutinationstheorie

- Verbal endings from personal promouns (e.g. first singular **-mi* cf. 1 person pronoun **me*)
- Verbal tenses from fusion with auxilialries (e.g. Germanic weak preterite in *–d* from the root **dhē* of the verb ‘do’)
- Nominative singular suffix *-s* traced back to the pronoun stem **so-* ‘he’
- Ancient Greek article < demonstrative
- Complementizer ‘that’ in the IE languages < a neuter demonstrative or relative pronoun in the accusative or some other dependent case

MORPHOLOGY

PIE: highly fusional

Agglutinationstheorie

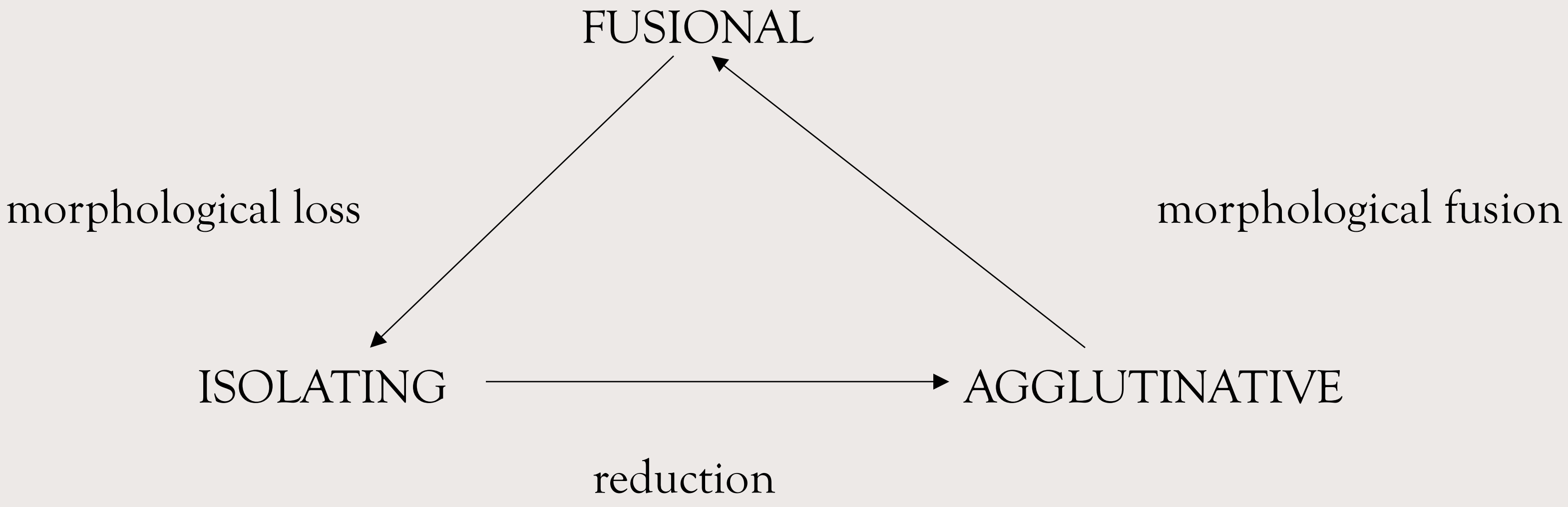
- Discarded by the Neogrammarians
- Revived by grammaticalization theory (Meillet 1912)

<u>Modern concept of grammaticalization</u>: Becoming more grammatical means losing in autonomy with respect to use by speakers. With respect to the language system, becoming more grammatical means getting increasingly subjected to the rules of grammar of a particular language. This brings with it reduction in semantic and phonological complexity, in syntagmatic variability and in morphological status (Lehmann 2016: 105)

MORPHOLOGY

Changes in morphological techniques

- From synthetic to analytic
- Grammaticalization: from analytic to synthetic



(Wahley 1997)

MORPHOLOGY

From fusional to agglutinative: Modern Armenian

Heřaxos (Telephone)		
Case	Singular	Plural
Nominative	heřaxos	heřaxos-ner
Accusative	heřaxos-ë	heřaxos-ner
Genitive	heřaxos-i	heřaxos-ner-i
Dative	heřaxos-in	heřaxos-ner-in
Ablative	heřaxos-ic’	heřaxos-ner-ic’
Instrumental	heřaxos-ov	heřaxos-ner-ov
Locative	heřaxos-owm	heřaxos-ner-owm

SYNTAX

Word order in PIE

Delbrück (1878, 1900): *traditionelle* vs. *occasionelle Wortstellung*

There is a traditional word order that is best recognised in the plain narrative. It is almost identical to the one we know from Latin. The subject begins the sentence, the verb closes it, the dative, accusative, etc. are placed in the middle, but in such a way that the accusative is immediately before the verb. The adjective comes before its substantive, as does the genitive.

…

The habitual sequence can be deviated from, if a term in the sentence needs to be emphasised, if the connection to another sentence requires a shift, or for whatever other reason. I call this deviant position the occasional one.

…

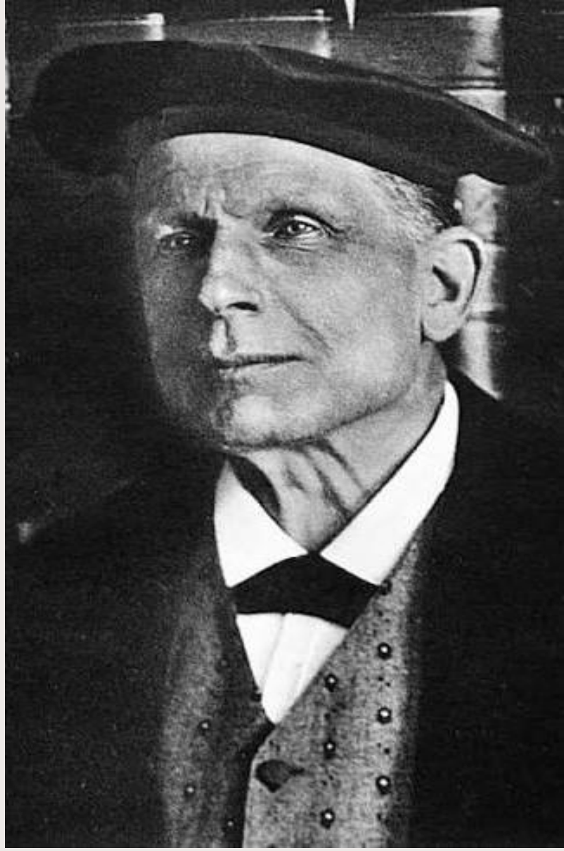
The occasional position of the words does not always result from a free momentary decision made by the speaker but can be influenced by tradition. Thus, as a basic law of occasional word order that runs through all Indo-European languages, it can be established that the word to be emphasized moves forward

SYNTAX

Wackernagel's Law

Jacob Wackernagel 1892 *Über ein Gesetz der indogermanischen Wortstellung* ('On a law about Indo-European word order')

→ Enclitic pronouns as well as enclitic and postpositive particles tend to be placed in second position in ancient Indo-European languages (Sanskrit, Ancient Greek, Latin)



ei	*kai*	*min*	*Olumpios*	*autòs*	*egeirei*
if	also	3SG.ACC	Olympian.NOM	self.NOM	push.PRS.3SG

'if even the Olympian himself pushes him', Homer *Il.* 13.58.

kai	*hē*	*gunḕ*	*eporãi*	*min*	*exiónta*
and	the	woman.NOM	see.PRS.3SG	3SG.ACC	go.out.PTCP.ACC

'and the woman saw him going out', Herodotus *Histories*, 1.10.2.

ho	*gár*	*toi*	*paîs*	*me*	*ho*
ART.NOM	PTC	PTC	child.NOM	1SG.ACC	ART.NOM
Sáturos	*apédra*				
Satyros.NOM	run_away.AOR.3SG				

'The boy Satyrus ran away form me.' (Pl. *Prot.* 310c3; Ancient Greek)

SYNTAX

1916 Hittite

(1) *piran=ma= at= mu* $^{\mathrm{mD}}$XXX.DU-*as* DUMU $^{\mathrm{m}}$*zida maniyahhiskit*
before CONN 3SG.ACC 1SG.OBL *A.*-NOM child *Z.* administrate.PRET.ITER.3SG
'before me Armadatta, the son of Zida, had administrated it';

(2) *n= as= mu= kan huwais*
CONN 3SG.NOM 1SG.OBL PTC escape. PRET. 3SG
'he ran away from me'.

nu= wa	*it*	*namma*	***apūn= ma=***	*wa=*	*tta*	
CONN QUOT	go.IMP.2SG	then	DEM.ACC=PTC=QUOT=2SG.DAT			
$^{\mathrm{LÚ}}$KÚ*R*	*Ḫayašan*	$^{\mathrm{D}}$U	*BELI-YA*		*karū*	
enemy	*H.*ACC	Stormgod	lord=1SG.POSS		already	
paiš	***nu=*** *war=*	*an=*	*kán*	*kueši*		
give.PST.3SG	CONN QUOT	3SG.ACC	PTC	strike.PRS.2SG		

'Go at last! The Stormgod my lord has already given you that Hasaya enemy and you will strike him.' (*KBo* 4.4 ii 56–57; Hittite)

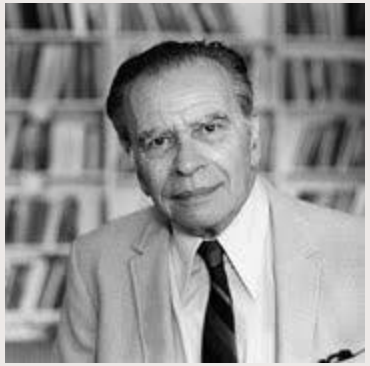
SYNTAX

Calvert Watkins "Preliminaries to a Historical and Comparative Analysis of the Syntax of the Old Irish Verb," *Celtica* 6 (1963) 1–49

(i) CONN (=clitici) ... V
(ii) X (=clitici) ... V
(iii) V (=clitici) ...
(iv) CONN (=clitici) V ...
(v) CONN (=clitici) ... PREV V
(vi) X (=clitici) ... PREV V
(vii) CONN (=clitici) PREV ... V
(Viii) PREV (=clitici) ...V
(ix) PREV (=clitici) V ...

SYNTAX

Word orde typology: basic order of meaningful elements



1. VSO (NA, NG, PrepN, NRel)
2. SVO (NA, NG, PrepN, NRel)
3. SOV (AN, GN, NPostp, RelN)

(Greenberg 1963)

→ What was the type of PIE?

SYNTAX

- W. P. Lehmann. *Proto-Indo-European Syntax* (1974) → PIE: rigid SOV

→ main support to reconstruction: typological comparison with Japanese

- Paul Friedrich (1975) *Proto-Indo-European syntax: the order of meaningful elements* → PIE: SVO
- R.M.R. Hall and Beatrice L. Hall (1971) *Underlying VSO word order in Indo-European* → PIE: VSO

- Gay Miller (1975) Indo-European: VSO, SOV, SVO, or all three?

SYNTAX

Reception:

- W.P. Lehmann’s Syntax → virtually disappeared from references

(essentially based on cross-linguistic typological comparison under the assumption of typological consistency)

- Delbrück, Wackernagel, Watkins → still going strong

(based on comparison of data from IE languages)

GRAMMATICAL CATEGORIES

Categories of nuons

Gender (not inflectional!)

Number

Case

Categories of Verbs

Aspect

Tense

Mood

Voice

GRAMMATICAL CATEGORIES

Inflection and derivation: Gender and number

inflection	derivation
no change of word class	change of word class
obligatory	non-obligatory
syntactic function	no syntactic function

Derivation serves the lexicon:

- *by creating new words*
- *by motivating the lexicon*

Different denotations → different words

Number:

Dog

Dogs

Gender:

magister / *magistra* (Lat); *Freund* / *Freundin* (Ger.)

GRAMMATICAL CATEGORIES

Inflection and derivation

"If the two inflexional forms differ semantically only, like Latin *urbs* (singular): *urbes* (plural), the status of such a pair will be intermediate between the relation basic word:derivative and the relation *urbs: urbem*. [...] *urbs:urbanus* = two different words; *urbs:urbes* = one word, with forms semantically different and having secondary syntactical functions; *urbs:urbem* = one word, with forms semantically identical, syntactically different" (Kuryłowicz 1964: 17)

Figure 1. IE nominal categories—inflection and derivation.

N → N → no change of class
vṛkaḥ → *vṛkīḥ*

Figure 2. Gender as non-prototypical derivation.

GRAMMATICAL CATEGORIES

Inflection and derivation

Similarities between gender shift and other types of derivation

Table 2. Some denominal nouns in the Indo-European languages.

diminutives	*putraḥ* 'son' → *putrakaḥ* (Skr.) *funis* 'rope' → *funiculus* (Lat.) ***Hund* (*m.*)** 'dog' → ***Hündchen* (*nt.*)** (Germ.) *gatto* 'cat' → *gattino* (It.) ***donna* (*f.*)** 'woman' → ***donnino* (*m.*)** (It.)
augmentatives	***palla* (*f.*)** 'ball' → ***pallone* (*m.*)** (It.)
agent nouns	***mercatus*** 'market' → ***mercator* (*m., f.: mercatrix*)** 'merchant' (Lat.)[3] ***Kunst*** 'art' → ***Künstler* (*m., f.: Künstlerin*)** 'artist' (Germ.) *science* → *scientist* (Engl.)
names of place	*granum* 'grain' → *granarius* 'granary' (Lat.) *acqua* 'water' → *acquaio* 'sink' (It.)
names of trees	***malum* (*n.*)** 'apple' → ***malus* (*f.*)** 'apple tree' (Lat.) ***mela* (*f.*)** 'apple' → ***melo* (*m.*)** 'apple tree' (It.) ***banana* (*f.*)** 'banana' → ***banano* (*m.*)** 'banana tree' (It.) *cerise* 'cherry' → *cerisier* 'cherry tree' (Fr.) *pomme* 'apple' → *pommier* 'apple tree' (Fr.)
abstract nouns	*phúsis* 'nature' → *phusikḗ* 'physics, science of nature' (Gr.) *lingua* 'language' → *linguistica* 'linguistics' (It.)
collectives	*Berg* 'mountain' → *Gebirge* 'mountains' (Germ.) ***frutto* (*m. sg.*)** 'piece of fruit' → ***frutta* (*f. sg.*)** 'fruit' (It.) ***muro* (*m. sg.*)** 'wall' → ***mura* (*f. pl.*)** 'city walls' (It.)

GRAMMATICAL CATEGORIES

What is gender?

“Genders are classes of nouns reflected in the behavior of associated words” (Hockett 1958: 231).

→ **Defining property of gender: setting up agreement classes**

“NOUN CLASSES and GENDERS are noun categorization devices realized outside the noun itself within a head-modifier noun phrase.” (Aikhenvald 2000: 17)

→ **Classificatory function of gender** (similar to other types of classifiers)

Non-classificatory function of gender

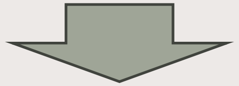
“It is a mistake to think of gender systems as systems for classifying things: to the extent that they do so it is secondary to their function to make it easier to keep track of links between constituents.” Dahl (2000: 113)

“The reason that gender is assigned to a noun is to enable anaphoric reference to that noun in discourse and deictic reference to that noun in the environment of speech.” (Frajzyngier and Shay 2003)

Grammatical Categories

Gender

- Ancient Indo-European languages → sex-based three-gender system (masculine, feminine and neuter).
- Brugmann (1891) → late development from an earlier animacy-based two-gender system (animate, morphologicaly corresponding to masculine, inanimate, correposnding to neuter)



Feminine gender formed through the addition of a special suffix (Brugmann -*ā, currently -*h2)

- Brugmann's explanation: Suffix originally abstract → the word for 'woman, *g'venâ (*gwenh?), was in origin an abstract noun from an otherwise unattested verb 'give birth', which caused the suffix to be re-interpreted as gender marker

GRAMMATICAL CATEGORIES

The Anatolian gender system

Two-gender distinction: non-neuter vs. neuter both in nouns and in pronouns

morphology:		SINGULAR			PLURAL	
		ANIMATE	INANIMATE		ANIMATE	INANIMATE
	nom.	-(a)s	Ø / -am		count	collective
	acc.	-(a)m	Ø / -am			(cf. verbal agreement)
all other cases		same endings		nom.	-es	-a
				acc.	-(a)nt	-a

dat/loc, genitive --> same ending for both genders

instrumental, ablative --> no plural

GRAMMATICAL CATEGORIES

Gender

SAME SUFFIX ALSO = NOMINATIVE/ACCUSATIVE NEUTER PLURAL!

- Extension to collective must have preceded extension to feminine, in order to accommodate the Anatolian data.
  → Influence of some feminine collective:
- Tichy (1993) extension to feminine started with the word *h -,*widhéweh* an ancient collective, indicating the relatives of a dead person, and later 'widow'
- !! NO traces of feminine collectives are attested anywhere in the IndoEuropean languages

GRAMMATICAL CATEGORIES

Gender

PROBLEM: nominative/accusative plural neuter = inflectional ending / feminine suffix = derivational

If we accept Tichy's theory the suffix underwent the following stages:

derivational suffix (abstract) → inflectional ending (neuter 'collective' plural) → derivational suffix (collective feminine) → stem vowel (–a stems: marker of an inflectional class)

Highly unlikely!!

Grammatical Categories

Gender

Table 1. Development of Suffix *-h_2

1.	derivational suffix (non-obligatory)		
2.	a.	neuter nouns:	
		inflectional suffix (nominative/accusative plural, obligatory)	
	b.	i	-*ā*- stems:
			marker of inflectional class ('theme vowel', obligatory)
		ii	first class adjectives:
			marker of inflectional class and feminine gender (obligatory)

Stages 1 and 2 chronologically ordered, stages 2a and 2b two separate developments:

2. a. a derivational suffix turns into an inflectional one, preserving (part of) its meaning;

2. b. a non-obligatory, meaningful suffix turns into a theme vowel, i.e. a purely grammatical, obligatory item, which is also interpreted as the marker of a noun class (i.e. of a grammatical gender).

GRAMMATICAL CATEGORIES

Gender

A seldom asked question: how did the two gender system come about?

(3) STAGES IN THE RISE OF GENDER MARKERS

generic nouns --> classifiers --> pronominal demonstratives --> attributive demonstratives --> determiners --> agreement markers

Gender:

Possible origins of gender:

Greenberg 1978, followed by Corbett (1991) Aikhenvald (2000)

Greenberg (1978: 79, followed by Corbett 1991: 313): "[t]he way in which gender arises need not be the same as that by which the system can expand by the development of new genders". → exponents of other grammatical categories become gender markers limited to the creation of new genders: "[g]ender systems may expand by adding new genders; this is generally done using existing morphological material." (e.g. rise of animacy based sub-genders for masculine nouns in the Slavonic languages)

GRAMMATICAL CATEGORIES

Gender

2 GENDERS - ANIMATE / INANIMATE (Anatolian)

	SINGULAR		PLURAL		
	ANIMATE	INANIMATE	ANIMATE	INANIMATE	
nom.	-(o)s	Ø / -om	count	stage (a_i) = Pre-PIE	no plural
acc.	-(o)m	Ø / -om		stage (a_{ii}) = PIE	collective (< abstract)
all other cases	same endings				

	ANIMATE	INANIMATE
nom.	-es	$-h_2 > -a$
acc.	-(o)ns	$-h_2 > -a$
all other cases	same form	

Figure 1 The Indo-European two gender system and agreement

DIFFERENCES IN AGREEMENT RESULT FROM DIFFERENTIAL MARKING OF CORE ARGUMENTS!!

GRAMMATICAL CATEGORIES

Origins of noun classes/gender systems

✓Increasingly grammaticalized systems of classifiers may develop into agreement systems (e.g. Bantu languages)

✓Other types of morpheme are re-interpreted as gender markers (e.g. Russian)

ORIGINS OF NOUN CLASSES/GENDER SYSTEMS

Gender from above vs. Gender from below

Gender from above: creation of new gender markers from an earlier system of classification. Usually involved in the rise of a gender system in languages without gender.

Stages in the rise of gender markers (grammaticalization):

Generic nouns → classifiers → pronominal demonstratives → attributive **demonstratives** → **determiners** → agreement markers

ORIGINS OF NOUN CLASSES/GENDER SYSTEMS

Gender from above vs. Gender from below

Gender from above:

Agreement triggering gender markers derive from semantically motivated non-agreement triggering markers, which had the function of singling out groups of nouns, either at the lexical level (derivational affixes) or at the syntactic level (classifiers). (Luraghi 2009). Fodor (1959) labels this process as “lexico-semantic”.

Origins of Noun Classes/Gender Systems

Gender from above vs. Gender from below

Gender from above:

- The classificatory function of gender is prominent
- High number of genders
- Overt gender markers

ORIGINS OF NOUN CLASSES/GENDER SYSTEMS

Gender from above vs. Gender from below

Gender from below:

Gender arises from special patterns of case marking. Such a process normally takes place in languages that already have a gender system. No new gender markers are created. Fodor (1959): morphologically or syntactically motivated process.

ORIGINS OF NOUN CLASSES/GENDER SYSTEMS

Gender from above vs. Gender from below

Gender from below:

Gender systems that derive from special agreement patterns are identified by agreement: hence, their primary function is to track discourse referents, that is, to indicate, through agreement, co-referring items, such as nouns and modifying attributes or nouns and anaphoric/deictic pronouns.

ORIGINS OF NOUN CLASSES/GENDER SYSTEMS

Gender from above vs. Gender from below

Gender from below:

- The tracking function is prominent
- Small number of genders.
- No overt gender markers.

ORIGINS OF NOUN CLASSES/GENDER SYSTEMS

Gender from above vs. Gender from below

NB: The two functions of gender coexist in all gender systems, but either one is maximized, partly depending on the origin of a system.

GRAMMATICAL CATEGORIES

Areally related gender systems (from Matasović 2004 and WALS):

- SUMERIAN: 2 genders, animate (socially active beings, some personified inanimate) / inanimate (including animals, abstractions, objects, collectives)
- ELAMITE: 2 genders , animate / inanimate (including collective plurals of animate nouns)
- DRAVIDIAN:
- Kannada, Tamil Gondi, Telugu: 3 genders, masculine / feminine / residue;
- Kolami, Parij Ollari: 2 genders, masculine / feminine (including inanimate);
- NW CAUCASIAN: no gender; 3 genders, male human / female human / inanimate;
- NE CAUCASIAN: 3, 4 and 5+ genders; all sex based (systems usually include a gender for animals; Tabasaran: 2 genders, animate / inanimate);
- BURUSHASKI: 4 genders, masculine / feminine / animals (including some highly individuated objects) / neuter (including objects, mass and abstractions);
- AFRO-ASIATIC: 2 genders, masculine and feminine (including singulative, abstractions, diminutives, socially inactive beings).

NON-CONFIGURATIONALITY IN ANCIENT IE LANGUAGES

Ken Hale (1983)

Warlpiri → absence of hierarchical relations among constituents, notably the non-existence of the VP.

Non-configurational languages → "flat" structure / or hierarchical structure at the Lexical Structure only (does not project on Phrase Structure).

WARLPIRI: Pama-Nyungan language spoken in the Northern Territory, Australia, by over 3000 people

NON-CONFIGURATIONALITY IN ANCIENT IE LANGUAGES

FEATURES OF NON-CONFIGURATIONALITY:

i. free word order

ii. referential null objects

iii. discontinuous constituents

IMPLICATIONAL SCALE:

discontinuous constituents > null objects > free word order (Baker 2001)

Jirwali Mohawk German

NON-CONFIGURATIONALITY IN ANCIENT IE LANGUAGES

(1) *Free word order in Warlpiri*

a.

Ngarrka-ngku	ka	wawirri	panti-rni
man-Erg	PresImpf	kangaroo	spear-Npst

"The man is spearing the kangaroo"

b. Wawirri ka pantirni ngarrkangku
Pantirni ka ngarrkangku wawirri
Ngarrkangku ka pantirni wawirri
Pantirni ka wawirri ngarrkangku
Wawirri ka ngarrkangku pantirni (Hale 1983:3)

NON-CONFIGURATIONALITY IN ANCIENT IE LANGUAGES

(2) *Discontinuous expressions in Warlpiri*

Maliki-rli-ji	yarlku-rnu	wiri-ngki
dog-Erg-1sgObj	bite-Pst	big-Erg

"A big dog bit me." (Hale et al 1995:1434)

(3) *Null anaphora in Warlpiri*

Purra-nja-rla	nga-rnu
cook-Inf-PriorC	eat-Pst

"Having cooked (it), (he/she/it) ate (it)." (Laughren 1989:326)

NON-CONFIGURATIONALITY IN ANCIENT IE LANGUAGES

(A) HEAD MARKING (MOHAWK):

(a) head marking (Mohawk):

Shako-núhwe's (ne owirá'a).

m.sgS/3plO-like-HAB NE baby

He likes them (babies). (Baker 1996)

(Features i, ii)

(B) DEPENDENT MARKING (JIWARLI):

(b) dependent marking (Jiwarli):

Juru-ngku ngatha-nha kulypa-jipa-rninyja parna.

sun-ERG 1sg-ACC be_sore-TRANS-PST head

The sun made my head sore. (Austin 1993)

(Features i,ii, iii)

NON-CONFIGURATIONALITY IN ANCIENT IE LANGUAGES

(A), (B)

Two types of non-configurationality

→ What is a configurational language? Best example: English → Word Order assigns grammatical relations (subject and object) and indicates constituency (noun modifier order cannot be changed)

Free occurrence of null anaphora lies at the heart of non-configurationality; its appearance draws a line between free constituent order, of the type known from German or Spanish, and “real” non-configurationality (Baker 2001a: 1437)

 What is «real» non-configurationality?

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Devine and Stephens (1999: 143-148)

Discontinuous constituents in Ancient Greek:

- survey of some features of non-configurationality
- attempt an explanation for the co-occurrence of such features
- state of affairs displayed by Ancient Greek indicative of ongoing change from non-configurationality to increasing configurationality

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Free word order and discontinuous constituents in Herodotus (Dover 1960, Devine & Stephens 2000)

(3) a. *apodexamenoi erga megala*

'having-performed deeds great' (Her 7.139);

b. *alia te megala erga apedexanto*

'other and great deeds they-performed' (Her 8.17);

c. *alia apodexamenos megala erga*

'other having-performed great deeds' (Her 1.59).

(4) *tauta es tous pantas Hellenas aperripse ho Kuros ta epea*

'these to the all Greeks directed the Cyrus the words' (Her 1.153).

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Discontinuous constituents in Latin

(5) *Arma* *virumque* *cano,* ***Troiae*** *qui* *primus*

arm(N).ACC.PL *man(M).ACC.SG+and* *sing.PRS.1SG* *Troy(F).GEN* *REL.NOM.SG.M* *first.NOM.SG.M*

ab ***oris*** *Italiam,* *fato* *profugus,*

from shore(N).ABL.PL *Italy(F).ACC* *destiny(N).ABL.SG* *fugitive.NOM.SG.M*

Lavinia*que* *venit* ***litora***

Lavinian.ACC.PL.N+and *come.PRF.3SG* *strand(N).ACC.PL*

"I sing the arms and the man, who, exiled by destiny, first came from the Trojan shores to Italy and to the Lavinian strand." Verg. Aen. 1.1-3 (Latin).

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Hewson, Bubenik (2006)

Emergence of configurationality in all Indo-European languages

→first detectable in the rise of adpositional phrases

→diachronically oriented (but it virtually only deals with increasing grammaticalization of adpositional phrases).

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Schäufele (1990)

Non-configurationality in Vedic Sanskrit

Reinhöl (2016)

Rise of configurationality in Indo-Aryan from Vedic Sanskrit onward

Configurationality → *by-product of the grammaticalization of certain function words* → *rise of postpositional phrase*

Old Icelandic

Farlund (1990), Sigurðsson (1993), Rögnvaldsson (1995)

→ also deal with null objects

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Jiwarli-like properies of Ancient Indo-European languages

NPs are case marked and the verb lacks agreement morphemes: thus, grammatical relations are marked on the NPs (i.e. on the dependents). Both the order of constituents and the order of words within constituents are free (that is, constituents can be discontinuous), and null anaphora is extensively used for subjects and objects

→ *Verby vs. nouny adjectives*

"discontinuous constituents are possible only in languages with no more than a weak N/A contrast" Baker (2001a: 1437) --> adjectives are predicated of nouns, rather then be dependents.

"L'adjectif n'est nullement lié au substantif. Il est généralement au même cas, au même nombre, et, ce qui est le trait caractéristique de l'adjectif, au même genre ..., mais parce qu'il s'applique au même objet." Meillet and Vendryes (1924: 530)

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Adjectives

tābhih	*jvalantībhih*	*dīpyamānābhih*	*upauteti*	*rājānam*
DEM.INS.PL	flaming.PTCP.INS.PL	shining.PTCP.INS.PL	approach.PRS.3SG	king(M).ACC.SG

"With the flaming, shining ones (sc. weapons) he approaches the king." AB 8.24.6 (Vedic Sanskrit).

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Adjectives

Indian grammarians found it difficult to distinguish between *visesana* 'qualifier' and *visesya* 'qualified' in a noun-adjective construction independently of the meaning intended by speakers in each given context. The 6^{th}-7^{th} century grammarian and philosopher Bhartṛhari, for example, "maintains that the … two terms … represent syntactic categories …; they refer to a word as a member of a combination and not as an isolated individual." Bath (1994: 170-171)

Greek adjectives with no gender distinction --> properties which are usually predicated of human beings *pénēs* 'poor', *Héllēn* 'Greek', *phugás* 'fugitive'. some of these adjectives always refer to human males, while some others always refer to human females: thus, their categorial status seems closer to nouns (they have inherent gender) Kühner, Blass (1890: 547-51)

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

Referential null objects

sadyó jātá *óṣadhībhir* *vavakṣe*

just born.PTCP.PRF.NOM.SG.M plant(F).INS.PL grow.PRF.MID.3SG

yádī várdhanti *prasvò* *ghṛténa*

when increase.PRS.3PL shoot.NOM.PL.F clarified.butter(N).INS.SG

"Just born, (Agni) has grown by means of the plants, when the shoots increase (him) with clarified butter." RV 3.5.8ab (Vedic Sanskrit);

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

ou gàr oímai themitòn eînai ***ameínoni*** ***andrì*** *hupò*

NEG PTC think.PRS.1SG righteous.ACC.SG.N be.INF better.DAT.SG.M man.DAT.SG.M under

kheíronos bláptesthai. apokteíneie mentàn ísōs ḕ exeláseien ḕ

worse.GEN.SG.M injure.INF.M/P kill.OPT.PRS.3SG PTC equally or banish.OPT.PRS.3.SG or

atimṓseien

disfranchise.OPT.PRS.3SG

“For I believe it’s not God’s will that a better man be injured by a worse. He might however perhaps kill (him), or banish (him), or disfranchise (him).” Pl. Apol. 30d (Ancient Greek);

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

dverginn mælti, at sá baugr skyldi vera

dwarf say.PRF.3SG that DEM.NOM.SG.M ring(M) should.PRF.3SG be.INF

hverjum hofuðsbani, er atti

whosoever.DAT.SG death REL have.PRF.3SG

“The dwarf said that that ring should bring death to anybody who possessed (it)” (Old Icelandic, from Sigurðsson, 1993, p. 248).

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

"Un verb indo-européen ne 'gouvernait' pas le cas de son complément; mais le nom apposé au verbe se mettait au cas exigé par le sens qu'il exprimait lui-

même." Meillet and Vendryes (1924: 522) --> Valence was **semantic** not **syntactic**

Semantic valence --> number of participants which are typically involved in an event

Syntactic valence --> number of actual constituents which a verb needs in order to stand in a grammatically acceptable construction (see Payne 1997: 169-170 and Luraghi, Parodi 2008: 197-199).

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

quaero, ecquid litterarum. Negant. … confessi sunt se accepisse,

ask:prs1sg indef.nom.n letter:gen.pl.f deny:prs.3pl confess:prf.3pl refl.acc take:prf.inf

sed excidisse in via

but drop:prf.inf in road:abl.sg.f

“I ask (the servants) if they have found any letters. They say they haven’t. … they confessed they had taken some, but had lost them on their way” (Cic. Att. 2.8);

Caesar exercitum reduxit et in Aulercis

Caesar:nom army:acc.sg.m take.back:prf.3sg and in Aulercian:abl:pl.m

Lexoviisque, …, in hibernis conlocavit

Lexovian:abl.pl.m+and in winter.camp:abl.pl.n settle:prf.3sg

“Caesar took his soldiers back and let them settle in the winter camp among the Aulercians and the Lexovians” (Caes. Gal. 3.27);

NON-CONFIGURATIONALITY IN ANCIENT INDO-EUROPEAN LANGUAGES

LATE LATIN

multua pecora habet, multum lanae, caput praeterea

many:acc.pl.n cattle:acc.pl.n have:prs.3sg much wool:gen.sg.f head:acc.sg.n especially

durum

hard:acc.sg.n

"He has many heads of cattle, plenty of wool, an especially hard head." (Petr. 39).

et obtuli eum discipulis tuis et non potuerunt

and bring:prf.1sg 3sg.scc.m disciple:dat.pl.m poss.2sg.dat.pl and neg can:prf.3pl

curare eum

cure:inf 3sg.acc.m

"And I brought him to your disciples, but they could not cure him." (Mt. 17.16).

THE RISE OF CONFIGURATIONALITY

OLD ITALIAN

or non avestú la torta? Messer sí: ebbi

ptc neg have:prf.2sg+2sg.nom art.sg.f cake:sg.f sir yes have:prf.1sg

“So, did you have the cake? Yes sir, I did!” (Nov. 79).

la torta l’ ha mangiata tutta Giovanni

art.sg.f cake.sg.f 3sg.acc.f have:prs.3sg eat:ptcp.sg.f all:sg.f John

“John ate the whole cake, it was John who ate the whole cake.”

THE RISE OF CONFIGURATIONALITY

Data extraction from syntactically annotated corpora (dependency treebanks)

Table 1. Texts composing the treebanks

	Classical Greek	Late Greek	Classical Latin	Late Latin
Author	Herodotus	Septuagint	Caesar and Cicero	Jerome
Title	*Histories* (I.1–VII.83)	*New Testament* (Matthew I.1 – Acts of the Apostles V.10)	*The Gallic War* (I.1–VII.77) and *Letters to Atticus* (I.1–VI.9)	*Vulgate* (Genesis I.1 – Acts of the Apostles XIV.11)
Date	440–429 BCE	49–150 ca. CE	58–50 BCE and 68–43 BCE	382–413 CE

THE RISE OF CONFIGURATIONALITY

A network is a graph consisting of a set of nodes V and a set of edges E.

A network can be induced from a treebank by setting up an equivalence between

(i) nodes and properties of words (e.g., their form, lemma, part-of-speech tag, etc.)

(ii) edges and word relations (e.g., precedence in linear order, dependency in syntax, etc.).

(11) *Et* *vocabis* *nomen* *eius* *Iesum*
and call:FUT.2SG name:ACC 3SG.GEN Jesus:ACC
"And you will call him Jesus." (Matthew 1.21)

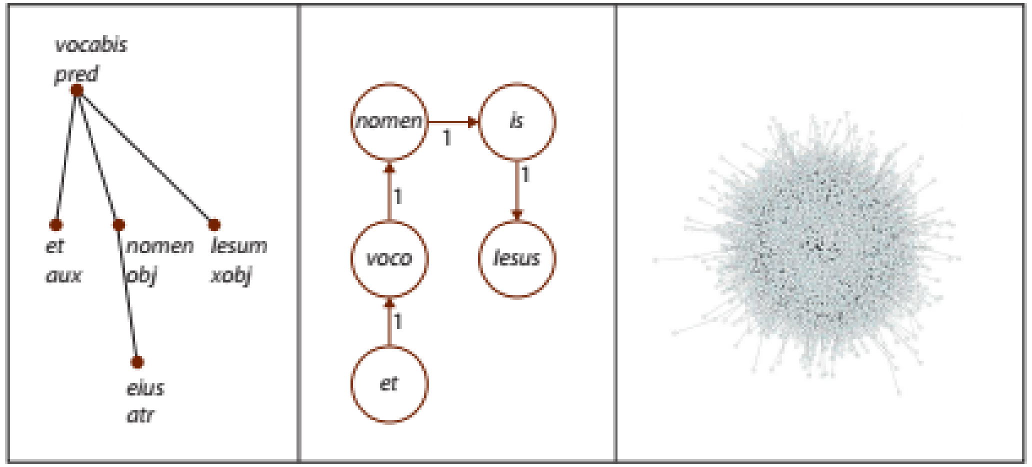


Figure 1. An example of syntactic tree (left), its equivalent network representation (center) and the global network resulting from a similar transformation of the whole treebank

THE RISE OF CONFIGURATIONALITY

FREE WORD ORDER

For Ancient Greek, we observe a shift from indifference regarding OV or VO order to a clear preference for VO. In Latin, the preference changes by swapping the order from OV to VO.

Table 3. Counts of objects following (VO) or preceding (OV) verbs in the four treebanks

	VO	OV
Classical Greek	3762	3409
Late Greek	4501	2158
Classical Latin	1473	5471
Late Latin	4884	2791

THE RISE OF CONFIGURATIONALITY

(12)	*Qui*	*diutissime*	*impuberes*	*permanserunt,*	*maximam*
	REL.NOM.PL	long	chaste:NOM.PL	remain:PRET.3PL	highest:ACC

	inter	*suos*	*ferunt*	*laudem*
	among	POSS.3PL.ACC.PL	carry:PRS.3PL	praise:ACC

"Those who have remained chaste for the longest time, receive the greatest commendation among their people." (Caes. G. 6.21)

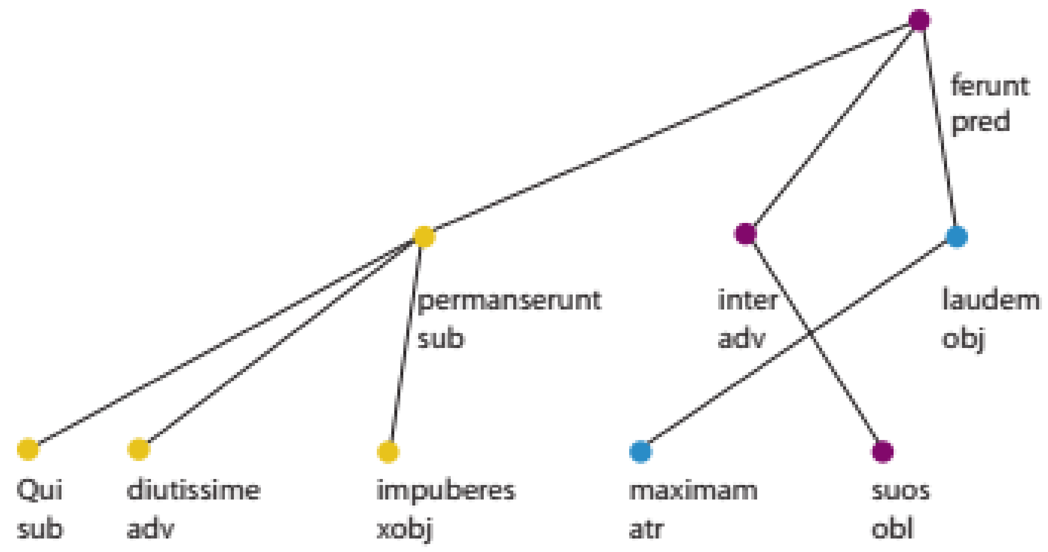


Figure 4. Discontinuous constituent in Classical Latin retrieved by the query

THE RISE OF CONFIGURATIONALITY

Discontinuous NPs

Table 5. Counts of pairs divided by at least a word in a gap (i.e., creating discontinuities) by kind of constituent

Language variety	Noun phrases (determiner)	Noun phrases (adjective)
Classical Greek	400	85 + 206 = 291
Late Greek	112	12 + 23 = 35
Classical Latin	–	221 + 58 = 279
Late Latin	–	25 + 18 = 43

THE RISE OF CONFIGURATIONALITY

To estimate the number of referential null objects, no direct source for information is available either in local or in global networks, since co-reference and implicit nodes are not present in all the treebanks.

Instead, this measure was approximated by the percentage of third person personal pronouns among the objects of verbs.

We assumed that, if the object is mandatory, it must surface as a pronoun in some constructions.

Table 7. Counts of the objects (total and subsets)

	CG	LG	CL	LL
objects	7171	6659	6944	7675
of which 3rd person personal pronouns	295 (4.11%)	789 (11.85)	167 (2.40)	763 (9.94)
of which in coordinated constructions	5 (0.07%)	58 (0.87%)	10 (0.14%)	57 (0.74%)

THE RISE OF CONFIGURATIONALITY

Third person pronouns (*autós* for Ancient Greek, *is* for Latin) climb the ranking in the late varieties compared to the classical varieties.

The evidence points towards a change in the role of third person pronouns

→ rising tendency to appear as verb objects points toward the overt realization of the object even in contexts in which its referent can be recovered from discourse.

Table 8. Ranking of the most connected nodes of global networks by number of edges

Position	CG	LG	CL	LT
1	*ho*	*ho*	*et*	*et*
2	*kaí*	*kaí*	*sum*	*sum*
3	*dé*	*autós*	*qui*	*is*
4	*eimí*	*dé*	*que*	*in*
5	*hoûtos*	*eimí*	*is*	*autem*
6	*autós*	*egṓ*	*ego*	*qui*

THE RISE OF CONFIGURATIONALITY

Free word order. Words in late varieties show a higher regularity in the co-occurrence patterns.

Constituents. The number of discontinuous NPs drops dramatically in late varieties.

Null referential direct objects. Third-person personal pronoun objects increase in number inside the treebanks, and third person pronouns increase in degree inside the global networks in late varieties.

IS THERE A CORRELATION?

THE RISE OF CONFIGURATIONALITY

Based on the figures in Table 9, we found a correlation to be statistically significant and strong.

Indeed, the count of third-person personal pronouns in coordination correlates negatively with the number of non-projective adjective-noun pairs.

Since we linked the former with the absence of the null anaphora of objects, it turns out that the occurrence of null referential direct objects and the occurrence of discontinuous NPs are interdependent.

Table 9. Pearson coefficients (bottom-left) and their significance expressed as p-value (top-right) of the correlations among the variables measured in both global and local networks

	Density	Discontinuous NPs w/ adjective	3rd person personal pronouns in coordination
Density	–	$p = 0.4692$	$p = 0.4884$
Discontinuous NPs w/ adjective	$\rho = -0.53$	–	$p = 0.0007$
3rd person personal pronouns in coordination	$\rho = 0.51$	$\rho = -0.99$	–

WHY DID CONFIGURATIONALITY ARISE?

Indo-European languages

SEMANTIC CONSTITUENCY TURNED INTO SYNTACTIC CONSTITUENCY IN THREE STEPS:

1. adpositional phrases
2. NPs with modifiers
3. VP → obligatoryconstituents order

WHY DID CONFIGURATIONALITY ARISE?

RISE OF RIGID WORD ORDER

Is loss of morphological case the cause?

Questionable: Rögnvaldsson (1995) remarks that Old Icelandic, which is considered non-configurational, has the same number of cases as Modern Icelandic, which is configurational

WHY DID CONFIGURATIONALITY ARISE?

DISAPPEARANCE OF DISCONTINUOS NPs

Baker (2001: 1437) → for discontinuous NPs to be allowed, a "particular kind of case marking is required". Herman (1985: 347) → indicating what lexemes belong to a certain constituent is a function of case systems in just the same way as indicating grammatical relations.

BUT: actual data clearly point toward an early development of configurationality within NPs!

WHY DID CONFIGURATIONALITY ARISE?

ADPOSITIONAL PHRASE

grammaticalization of adpositional phrases brought about increasing loss of independent semantic contribution of cases to the meaning of the construction → development preceded loss of the case system by many centuries (e.g. Latin).

DESEMANTICIZATION OF CASES IS A CONSEQUENCE, RATHER THAN A CAUSE OF THE RISE OF CONFIGURATIONALITY

WHY DID CONFIGURATIONALITY ARISE?

CONFIGURATIONALITY →

<u>BY-PRODUCT OF RELATEDNESS OF FREQUENTLY CO-OCCURING ITEMS</u>

1. spatial adverb specifying the precise spatial meaning of a NP inflected in a certain case → grammaticalization → <u>adpositional phrase</u>
2. nouns that indicated properties often accompanied other nouns → appositions, originally agreement in case → frequent co-occurrence with other nouns caused them to be felt as subordinate → agreement in gender → configurationality within NPs
3. frequent co-occurrence of certain verbs with NPs inflected in the accusative → transitivity, → direct objects obligatory even when they were not expressed though a NP → <u>rise of VP and loss of null objects</u>

CONSTRUCTIONALIZATION

Recall from previous lecture:

Traugott (2014)

constructionalization perspective: diachronic process understood as a change that <u>encompasses the whole construction</u>

CONSTRUCTIONALIZATION

CONSTRUCTIONALIZATION

1. from spatial adverb to adposition
   a) *spatial adverb adds some specification to NP which through case markig indicates spatial relation*
   b) *adpositional phrase: the adposition becomes the marker of the spatial relation, the case marker loses independent relevance*

CONSTRUCTIONALIZATION

CONSTRUCTIONALIZATION

2. loss of discontinuous NPs

   a) *two nouns refer to the same referent, they are in an appositional relation and are inflected in the same case because they fulfill the same semantic role*
   b) *the appostion becomes an attribute of the headnoun and develops gender agreement* → *headnoun: controller / attribute: target of agreement*

CONSTRUCTIONALIZATION

CONSTRUCTIONALIZATION

3. rise of VP

a) *verb only has semantic valency* → *direct objects recoverable from the contex are not overtly realized*

b) *grammaticalization of valency* → *direct objects become obligatory*

overtly realized direct objects become markers of transitivity

TOWARD HEAD-MARKING NON-CONFIGURATIONALITY

Loss of null objects in Late Latin

*et obtuli **<u>eum</u>** discipulis tuis et non potuerunt*

and bring:PRF.1SG 3SG.SCC.M disciple:DAT.PL.M POSS.2SG.DAT.PL and NEG can:PRF.3PL

*curare **<u>eum</u>***

cure:INF 3SG.ACC.M

“And I brought him to your disciples, but they could not cure him.” (Mt. 17.16).

TOWARD HEAD-MARKING NON-CONFIGURATIONALITY

VO vs. OV in Modern Italian

Maria ha mangiato la torta

Maria has eaten:M the cake

“Maria ate the cake”

la torta ***l’*** *ha mangiata Maria*

the cake(F) **it(CLITIC)** has eaten:F Maria

“Maria ate the cake, it was Maria who ate the cake.”

TOWARD HEAD-MARKING NON-CONFIGURATIONALITY

French: obligatory clitics subjects

moi je ne sais pas ce qu'il veut, ce garçon la

me I NEG know NEG DEM REL he wants DEM guy there

"I don't know what he wants, that guy."

TOWARD HEAD-MARKING NON-CONFIGURATIONALITY

Bossong (1998: 32) points out that a sentence such as:

il la voit, la femme

he her sees the woman

"He sees her, the woman."

tends to be realized without any intonational break:

il la voit la femme

"He sees the woman."

TOWARD HEAD-MARKING NON-CONFIGURATIONALITY

Bossong: spoken French moving in the direction of → doubling all arguments by means of clitics hosted by the verb

Il la lui a donné, son père, à Jean, sa moto

he it.F him has given his father to John his motorbike

"John's father gave him his motorbike" (Tesnière, 1959: 175).

TOWARD HEAD-MARKING NON-CONFIGURATIONALITY

clitic doubling makes possible binding of the possessive *son* with the oblique NP *à Jean* → impossible with normal word order and normal intonation:

**Son*$_i$ *père a donné à Jean*$_i$ *sa moto.*

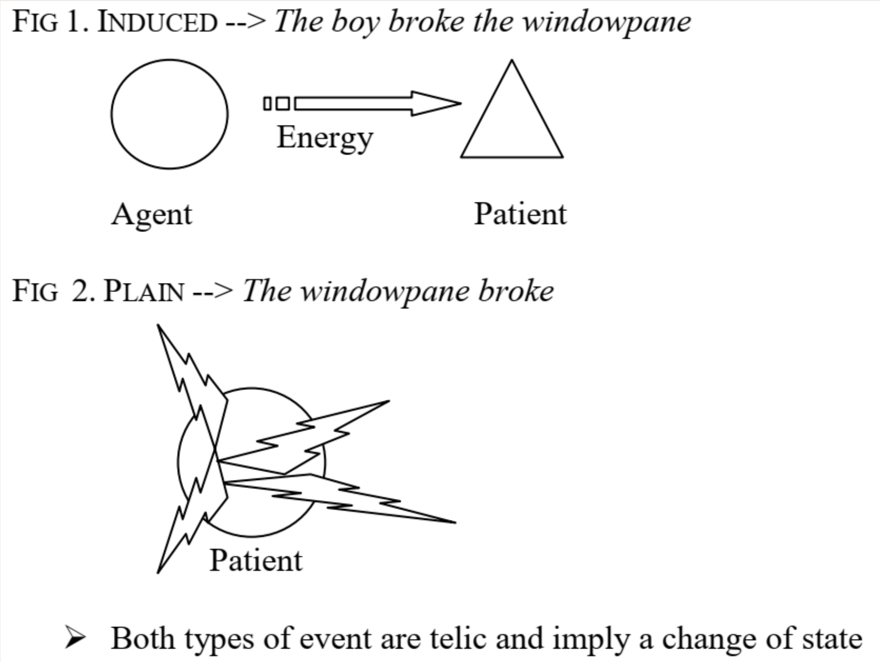
"His father has given John his motorbike." (ungrammatical with his = Jean's)

→ pattern typical of non-configurational languages of the Mohawk type, i.e. head marking ones

BASIC VALENCY

Nichols et al. (2004) "…**basic valence orientation** … serves as a useful building block for characterizing the **lexical and morphosyntactic signatures of languages, language areas, and language families**." (Nichols et al. 2004: 149)

Transitivity --> semantic rather than syntactic concept (Hopper, Thompson 1980; Nichols et al. 2004: 150): *eat* „intransitive". Better defintion: plain vs. induced --> Causative alternation (cf. Haspelmath (1987, 1993)



Basic valency

Pair	Plain	Induced
1	'laugh'	'make laugh, amuse, strike as funny'
2	'die'	'kill'
3	'sit'	'seat, have sit, make sit'
4	'eat'	'feed, give food'
5	'learn, know'	'teach'
6	'see'	'show'
7	'be/become angry'	'anger, make angry'
8	'fear, be afraid'	'frighten, scare'
9	'hid, go into hiding'	'hide, conceal, put into hiding'
10	'(come to) boil'	'(bring to) boil'
11	'burn, catch fire'	'burn, set fire'
12	'break'	'break'
13	'open'	'open'
14	'dry'	'make dry'
15	'be/become straight'	'straighten, make straight'
16	'hang'	'hang (up)'
17	'turn over'	'turn over'
18	'fall'	'drop, let fall'

BASIC VALENCY

Group I (oriented)

(a) **transitivizing** vs. (b) **intransitivizing languages**:

(a) **Basic form is intransitive**: transitive form is morphologically more complex

(special case: the basic form is an adjective) → **Augmented correspondences**: (Nanai, Tungusic)

(b) **Basic form is transitive**: intransitive form is morphologically more complex → **Reduced correspondences**: (Russian, Indo-European)

Basic valency

…

(a) Augmented correspondences: (Nanai, Tungusic)
(b) Reduced correspondences: (Russian, Indo-European)

TAB. 1: INTRANSITIVE / TRANSITIVE VERBS IN NANAI UND RUSSIAN

INTRANSITIVE:	‘learn’	‘fear’	‘hide’(go into hiding)
TRANSITIVE:	‘teach’	‘frighten, scare’	‘hide’ (put into hiding)
Nanai	*otoli-*	*mian-*	*siri-*
	otoli-wa:n-	*mian-bo-*	*djaja-*
Russian	*učit’-sja*	*bojat’-sja*	*prjatat’-sja*
	učit’	*pugat’*	*prjatat’*

(adapted from Nichols et al. 2004: 151)

BASIC VALENCY

Group II (non-oriented)

(c) **Undetermined correspondence** (both members of the pair treated as derived):

- <u>Suppletion</u> (*sterben/töten*)
- <u>Ambivalent</u> (labile) (*brechen*)
- (Conjugation class change)

Conjugation change: “In languages which classify verbs by conjugation class (i.e., have sets of allomorphs for inflectional paradigms; we also code lexically determined theme vowels, extensions, and the like as differences of conjugation class), and conjugation classes are inflectional classification rather than derivational categories, verbs showing this correspondence belong to different conjugation classes in their intransitive and transitive uses but are otherwise underived.” (Nichols et al. 2004: 159)

→ Only compelling example: West Armenian

BASIC VALENCY

Group II (non-oriented)

(d) **Neutral correspondence:**

- Ablaut (*fallen/fällen*) (→ but cf. Plank & Lahiri 2015)
- Double derivation (equipollent)

Yupik (Siberian)	**aghagh-nga**	**aghagh-te-**
	'hang'	'hang' (tr.)

- Auxiliary / light verb change

Ingush **(East Caucasian)**	**ieghaz-v.uoda (angry-go)** **ieghaz-v.ug (angry-lead)**	**'be/get angry'** **'anger, make angry'**
Hausa (Chadic)	yi dariya (lit. 'do laughter') ba dariya (lit. 'give laughter')	'laugh' 'make laugh'

BASIC VALENCY AND VALENCY ORIENTATION
FREQUENCIES (FROM NICHOLS ET AL 2004)

- **Universal preferred lexicalization** Nichols et al.'s (2004: 172), cf. Haspelmath (1993):

"Other things being equal, languages prefer to lexicalize as basic verbs that typically select animate subjects"

	ANIMATE	INANIMATE
Augmented	285.5	203.5
Reduced	42.5	106.5
Double	61	75
Ambitransitive	28.5	66.5
Conjugation	0.5	7
Auxiliary change	14	25
Ablaut	22	30.5
Suppletion	206	98
Adjective	3.5	5

Basic Valency

… Indo-European languages in Nichol et. al's sample (out of 80 total languages):

Russian	Detransitivizing --> reflexive middle
German	Detransitivizing --> reflexive middle
Modern Greek	Detransitivizing/Undetermined (suppletion, lability)
West Armenian	Transitivizing
Hindi	Transitivizing
Portuguese	Undetermined (suppletion)
Ossetian	Undetermined (lability)/Transitivizing

BASIC VALENCY

BASIC VALENCY ORIENTATION IN MODERN HIGH GERMAN

Plank&Lahiri 2015: 6)

"there are sometimes other formal means to distinguish intransitives and transitives even for the same pairs, such as alternations of stem vowels (e.g., sitzen 'sit, be seated' – setzen 'set, seat',

futtern 'eat, stuff oneself' (colloquial) – füttern 'feed') which Nichols et al. do not code as being directed, and which would therefore have to be credited to the account of indeterminateness." (ib.)

Strong = basic and intransitive, weak = derived and transitive in German

	INTRANSITIVE	←	TRANSITIVE
a.	*sich setzen* 'take a seat'		*setzen* 'set'
	sich ärgern 'be/become angry'		*ärgern* 'annoy'
	sich amüsieren 'enjoy oneself'		*amüsieren* 'amuse'
	(sich) erschrecken 'fear, be frightened'		*erschrecken* 'frighten'
	sich verstecken 'hide oneself, go into hiding'		*verstecken* 'hide'
b.	*(sich) öffnen* 'become open'		*öffnen* 'cause to become open'
	(sich) (um-)drehen 'turn over/around'		*(um-)drehen* 'cause to turn over/around'

BASIC VALENCY

BASIC VALENCY ORIENTATION IN MODERN HIGH GERMAN

Strong = basic and intransitive, weak = derived and transitive in German

OR MAYBE NON-ORIENTED?

TAB. 4: BASIC VALENCY ORIENTATION IN MODERN HIGH GERMAN - NON-ORIENTED (?)

NON-CAUSATIVE	CAUSATIVE
essen 'eat'	*ätzen* 'feed' (of animals and their young), 'etch' (make eat into)
trinken 'drink'	*tränken* 'water' (horses, cattle)
er-saufen 'drown' (be killed)	*er-säufen* 'drown' (kill by submerging in water)
ge-nesen 'convalesce'	*nähren* 'nourish'
saugen 'suck'	*säugen* 'suckle'
beissen 'bite'	*beizen* 'marinate'
sitzen 'sit'	*setzen* 'set'
liegen 'lie'	*legen* 'lay'
hängen 'hang'	*henken* 'execute by hanging'
fahren 'drive, ride'	*führen* 'lead, drive'
springen 'spring, jump'	*sprengen* 'blow up'
dringen 'penetrate'	*drängen* 'urge, push, press'

Basic Valency

…

Tab. 5: Homeric Greek Verb pairs

Animate Verbs	plain	induced	strategy
(1) 'laugh'-'make laugh'	*geláō*	periphrasis	TRANSITIVIZING
(2) 'die'-'kill'	*thnḗskō*	*kteínō*	SUPPLETION
(3) 'sit'-'seat'	*hézomai*	-	AUGMENTATION
	hízomai	*hízō*	VOICE ALTERNATION
	hízō		LABILE ACTIVE
(4) 'eat'-'feed'	*esthíō*	*bóskō*	SUPPLETION
(5) 'learn'-'teach'	*daēmenai<*daō* *didáskomai*	*dédae* *didáskō*	AUGMENTATION VOICE ALTERNATION
(6) 'see'-'show'	*horáō*	*deíknumi*	SUPPLETION
(7) 'be angry'-'anger'	*kholoûmai*	*kholóō*	VOICE ALTERNATION
(8) 'fear'-'frighten'	*deídō*	*deidíssomai*	AUGMENTATION
(9) 'hid'-'hide'	*keúthomai*	*keúthō*	VOICE ALTERNATION
Inanimate Verbs			
(10) 'boil'-'boil'	*zéō*	-	-
(11) 'burn'-'burn'	*kaíomai*	*kaíō*	VOICE ALTERNATION
(12) 'break'-'break'	*rhḗgnumai*	*rhḗgnumi*	VOICE ALTERNATION
(13) 'open'-'open'	*oígnumai*	*oígō*	VOICE ALTERNATION
(14) 'be(come) dry'-'dry'	*térsomai*	*tersaínō*	AUGMENTATION- VOICE ALTERNATION
(15) 'be(come) long'-'lengthen'	*tanúomai*	*tanúō*	VOICE ALTERNATION
(16) 'hang'-'hang'	*krémamai*	*kremánnumi*	VOICE ALTERNATION
(17) 'turn over'-'turn over'	*stréphomai*	*stréphō*	VOICE ALTERNATION
(18) 'fall'-'drop'	*píptō*	*híēmi*	SUPPLETION

Basic Valency

Tab. 6: Hittite Verb pairs

		ANIMATE VERBS		
		MEANING	PLAIN	INDUCED
1.	?	laugh cry	*hahhars-* *wiya-*	- -
2.	Undetermined	die	*ak-*	*kuen-*
3.	Transitivizing Transitivizing d	sit sleep	*es-* *ses-*	*ases-, assessanu-* *sas(sa)nu-*
4.	Transitivizing Transitivizing	eat drink	*et-* *eku-*	*adanna pai-* *akuwanna pai-*
5.	?	learn	*istamas-* 'hear'	-
6.	Undetermined	see	*aus-/u-, sakuwaya-*	*tekkussa-, tekkussanu-*
7.	Transitivizing Transitivizing	be angry worry	*kartimmiya-* *lahlahhiya-*	*kartimmiyahh-* *lahlahhinu-*
8.	Transitivizing	fear	*nah-, nahsariya-*	*nahsarnu-*
9.	Undetermined	hide	*munnai-* (mid.)	*munnai-* (act.)
		INANIMATE VERBS		
		MEANING	PLAIN	INDUCED
10.	Transitivizing	boil	*ze-*	*zanu-*
11.	Transitivizing	burn	*war-*	*warnu-*
12.	Undetermined	break	*duwarna-* (mid.)	*duwarna* - (act.)
13.	Transitivizing	open	*hassanza* (adj.)	*hass-*
14.	Transitivizing /equip.	dry	*hat-, hatess-*	*hatnu-*
15.	Equipollent (Trans.)	become long	*dalukess-*	*daluganu-*
16.	Transitivizing	hang	*agank-*	*(anda) ganganu-*
17.	Transitivizing	turn over	*weh-*	*wahnu-*
18.	Undetermined	fall	*mauss-*	*pessiya-*

(from Luraghi 2012)

Tab. 7: Other animate verbs

	PLAIN	INDUCED
1.	*(anda)impai-* 'worry'	*(anda) aimpanu-* 'make worry'
2.	*hass-*, 'give birth'	*hassanu-* 'bring to birth'
3.	*hassikk-* 'be satiated'	*hassikkanu-* satiate
4.	*hatuk-* 'be fearsome'	*hatuganu-* 'terrify'
5.	*huis-*, 'live'	*huisnu* - 'rescue', 'heal' 'let live'
6.	*merr-* 'get lost', 'go missing'	*mernu-* (NH) 'cause to disappear, dissolve'
7.	*nink-* 'soak', 'get drunk'	*ninganu-* 'drench'
8.	*pukka-* 'be hateful'	*pugganu-* 'make hateful'
9.	*samen-* 'disappear'	*samenu-* 'make (something/-one) pass by, bypass, ignore (someone)'
10.	*tarra-* 'be able to'	*tarranu-* 'make powerful'
11.	*tariya* - 'be(come) tired'	*dariyanu* - 'make tired'
12.	*wak(ki)ssya-* 'omit', 'be absent'	*waggasnu-* 'omit'
13.	*werite-* 'fear'	*weritenu-* 'scare'

Tab. 8: Other inanimate verbs

	PLAIN	INDUCED
1.	*ars-* 'flow'	*arsanu-*, 'let flow'
2.	*hark-* 'perish'	*harganu-* 'destroy'
3.	*kist-* 'go out'	*kistanu-* 'put out'
4.	*lak-* 'be knocked down' (mid.)	*lak-* (act.) *laknu-* (NH)
5.	*lap-* 'glow'	*lapnu-* 'kindle'
6.	*mai-, miya-* 'grow', 'be born'	*miyanu-* 'let grow (vegetation)'
7.	*parkiya-* 'rise'	*parkiyanu-* 'make to rise'
8.	*samesiya-* 'burn (for fumigation) intr.' (mid.)	*samesiya-* 'burn (for fumigation) tr.' (act.) *samesanu-* 'burn' (only NH)
9.	*zappiya-* 'drop'	*zappanu-* 'let drop (liquid)'

Tab. 9: Other verb pairs with voice alternation --> mostly inanimate

		Meaning	Active	Middle
1.	*harp-*	split	transitive	intransitive
2.	*irha-*	finish	transitive	intransitive
3.	*lazziya-*	prosper, flourish/set straight, recover (only mid.)	transitive	intransitive from OH/OS
4.	*luluwai-*	survive/sustain	transitive from OH	intransitive from OH
5.	*marriya-*	melt down/melt	transitive from OH/NS	intransitive from OS
6.	*nai-*	turn	transitive from OS	intransitive from OS transitive NH
7.	*pars-*	break	transitive	transitive / intransitive from OH/NS
8.	*suwai-*	fill	transitive	intransitive
9.	*zinna-*	finish	transitive/ intransitive	intransitive

BASIC VALENCY

GOTHIC

No Middle voice, no reflexive middle (later development in Germanic languages), derived causative in -*ja*-, cf. Suzuki (1989), Ottosson (2013)

CAUSATIVE *JA*-VERBS FROM INACTIVE INTRANSITIVE VERBS (ANTICAUSATIVE CONTENT)

- ✓ *anþarans <u>ganasida</u>, ak sik silban ni mag ganasjan* 'he saved others, but he cannot save himself' Mark 15,31
- ✓ *hwas mag <u>ganisan</u>* 'who can be saved' Luke 18,26
- ✓ *<u>gadrausida</u> mahteigans af stolam* 'he pulled down princes from their thrones' Luke 1,52
- ✓ *ains ize ni <u>gadriusiþ</u> ana airþa inuh attins izwaris wiljan* 'not one of them falls to the ground without the will of the father' Matthew 10,29

BASIC VALENCY

TRANSITIVE-INTRANSITIVE (ANTICAUSATIVE) PAIRS

✓ Transitive weak *ja*-verb and *na*-verb derived from an adjective

fulljan 'fill' - *fullnan* 'become full' (*fulls* 'full'), *gabigjan* 'enrich' - *gabignan* 'be rich' (*gabigs* 'rich'), *managjan* 'increase' - *managnan* 'be plentiful, increase' (*manags* 'many'), *mikiljan* 'praise, magnify' - *mikilnan* 'be enlarged' (*mikils* 'big'), *usmerjan* 'spread abroad' - *usmernan* 'spread (intrans)' (cf. *wailamereis* 'admirable'), *gaqiujan* 'make alive' - *gaqiunan* 'become alive' (*qius* 'alive'), *dauþjan* 'kill' - *gadauþnan* 'die' (*dauþs* 'dead')

✓ Transitive weak *ja*-verb and *na*-verb without attested related adjective

drobjan 'stir up' - *drobnan* 'become restless', *usgaisjan* 'frighten' - *usgeisnan* 'be astonished', *afhwapjan* 'choke' - *afhwapnan* 'be choked', *fraqistjan* 'destroy' - *fraqistnan* 'be destroyed', *afslauþjan* 'perplex' - *afslauþnan* 'be amazed'

CHRONOLOGY OF THE CAUSATIVE ALTERNATION IN GERMANIC

(adpted from Ottorsson 2013)

- Causatives in *-ja-* attest to basic transitivizing orientation
- Development of *-na-* anticausatives in Proto-Germanic partly connected with loss of PIE inflectional middle
- Later development (not attested on Gothic): reflexive middle --> basic orientation changes to detransitivizing
- Development of labile verbs (non-oriented)

Basic Valency

…

Valency and lexical aspect

TAB. 11: MEDIA TANTUM

1.	*ā-*	'warm up'
2.	*ar-*	'stand'
3.	*es-*	'sit down'
4.	*iya-*	'walk'
5.	*isduwa-*,	'become known/apparent'
6.	*ki-*,	'lie'
7.	*kis-*,	'happen', 'become'
8.	*kist*-,	'go out, be extinguished'
9.	*pugga-,*	'be hateful';
10.	*tarra-*,	'be capable of'
11.	*tugga-*,	'be visible'
12.	*war-*,	'burn'
13.	*ze*-,	'be(come) cooked'

(from Luraghi 2012)

Stative: n. 1, 2, 4, 6, 10, 11, 13

BASIC VALENCY

Adjective, intransitive change-of-state (fientive), transitive induced (cf. Germ. verbs in *-ja-/-na-*)

suppiahh- 'purify' from *suppi-* 'pure', *suppess-* 'become pure'

idalawahh- 'harm, injure' from *idalu-* 'evil', *idalawess-* 'become evil'.

Stative verbs with *-ahh-* und *-ess-* correspondence: *nakke-* 'be important' vs. *nakkiyahh-* 'regard/ treat as important' and *nakkess-* 'become important, troublesome'.

Latin fientive derivates in *-ēsc-*: *rubēsco* 'I blush' from *ruber* 'red' *rubeō* 'I am red, blushing'; *senēscō* 'I grow old' from *senex* 'old' *seneō* 'I am old, have grown old' (Watkins 1971: Latin *-ēsc-* corresponds to Hittite *-ess-*)

Latin de-adjectival causative suffix *-ā- novāre* 'renew' from *novus* 'new', cf. Hittite *newahh-* 'renew'.

Are verbal roots unambiguosly specified for a single lexical aspect?

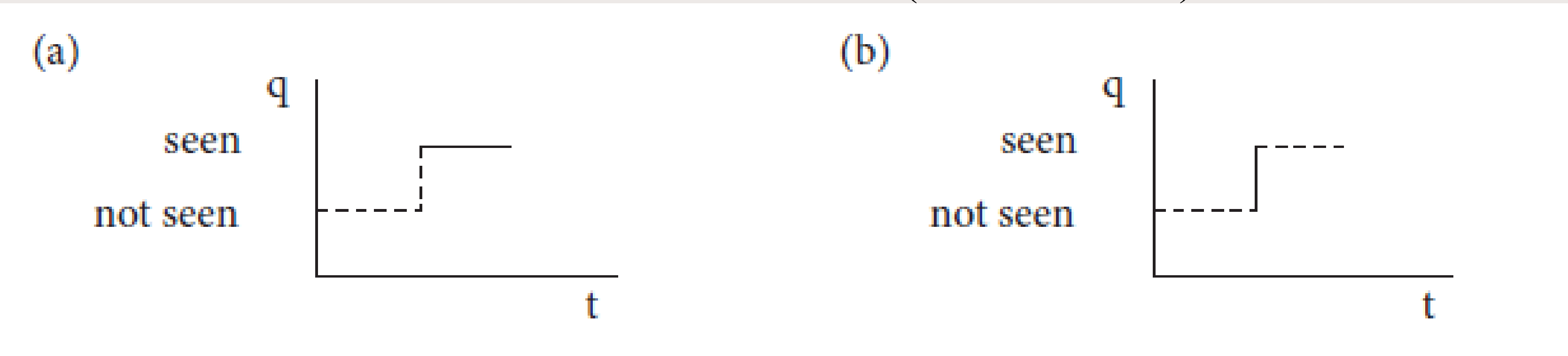
Basic valency

✓ Verbs can refer to events profiled in different ways within the same language

(a) *I **see** Mount Tamalpais*

(b) *I reached the crest of the hill and **saw** Mount Talmapais*

FIG. 3: ALTERNATIVE PROFILING FOR ENGLISH *SEE* (Croft 2012: 55)



✓ Cross-linguistic correspondence does not always hold

English *know* stative *vs.* Italian *conoscere* stative/change of state

--> *Ho conosciuto Giovanni un mese fa* ‘I met (*knew) John a month ago’

BASIC VALENCY

✓ Reference to state or change of state not always clear (examples from Ottosson 2013)

- State: *nu saiwala meina gadrobnoda*

  'Now my soul is troubled' John 12,27

- Change of state: *jah gadrobnoda Zakarias gasaihwands, jah agis disdraus ina*

  'the sight disturbed Zechariah and he was overcome with fear' Luke 1,12

- Unclear: *ni indrobnai izwar hairto*

  'do not let your hearts be troubled' John 14,1

BASIC VALENCY

Lexical apects and valency alternation

German: *sitzen / setzen* --> transitivizing

sich setzen / setzen --> detransitivizing

Homeric Greek: *hézomai* ‘be sitting’

hízomai ‘sit down’ / *hízō, eísa* (sigmatic aorist) ‘seat’

Slavic: **sed*- ‘sit down’

**sed-e-* ‘be sitting’ (stative) vs. **sad-i*- ‘let sit’ --> equipollent

(cf. Nichols 2006)

Russian: *sidet’* ‘be sitting’ (only imperfective)

sadit’-sja/sest’ ‘sit down’ / *sažat’/(u-, po-)sadit’* ‘seat’ --> detransitivizing

Plain = change of state root: induced adds causation

- Plain = stative root: induced adds change of state+causation (two steps)

Basic valency

Valency alternation and the middle voice

PIE *media tantum* (cf. Delbrück 1911):

- State
- Change of state

common feature: *lack of control*

▶ Lexical distribution?

Late development of voice opposition

Both directions

▶ Voice alternation: oriented or non-oriented?

<u>Morphological complexity</u>:

- older stage --> middle endings related only to perfect endings
- later stage --> middle endings related to active ending (cf. Gr. *-mai* etc.)

➢ Verbs denoting controlled actions cannot have stative counterparts --> voice opposition < canticaustive alternation with change of state verbs

Stative verbs --> only transtivizing strategies
Change of state verbs --> both transitivizing and detransitivizing strategies possible

BASIC VALENCY

Voice opposition originated out of anticausative alternation; reflexive meaning of the IE middle is a later development

THE MIDDLE VOICE IN VEDIC SANSKRIT

“The middle diathesis (also called ‘middle voice’) is usually said to function as a syncretic marker of several intransitive derivations: passive, anticausative (de-causative), reflexive, reciprocal; see examples below. This might indeed be thecase in Proto-Indo-European. However, one of the oldest documented Indo-Eu-ropean languages, Vedic Sanskrit, seems to attest the decay of the original system.” (Kulikov 2009: 76) [cf. Meiser 2010]

…

“The causative/anticausative distinction is the only valency-changing derivationwhich, unlike passive, reflexive and reciprocal, is quite regularly expressed bythe active/middle opposition, at least in early Vedic, as in med. *várdhate* ‘grows’ ~ act. *várdhati* ‘makes grow, increases’ or med. *réjate* ‘trembles’ ~ act. *réjati* ‘makes tremble’.

… in most cases, the middle type of inflexion is not the only marker of anticausative, being supported by the stem opposition … <u>transitive-causative presents with nasal affixes with the active inflexion are mostly opposed to middle thematic root presents</u> [emphasis added] … cf. *pávate* ‘becomes clean’~ *punā́ti* ‘makes clean’; *rī́yate* ‘flows, bubbles’~ *rinā́ti* ‘makes flow, makes bubble’.” (Kulikov 2009: 76, see also Kulikov 2000, 2013)

BASIC VALENCY

Innovation (Kulikov: degrammaticalization of the middle voice) or archaism?

Cf. Hittite (Hoffner&Melchert 2008):

reflexive --> mostly expressed by the particle *-za*

reciprocal --> mostly expressed by the numeral 'one', the demonstrative, or with the particle *-za*

passive --> often perophrastic (passive participle plus verb 'be'), passive meaning of middle rare and late

--> anticausative alternation

The prototype of the middle voice is based on the Greek middle --> Anatolian, Indo-Aryan more conservative